Pricing Spinny's Buyback Guarantee as a Real Option

MS5612 – Real Options Valuation

Group 4

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Outline

About Spinny

- Founded in 2015; operates in the **second-hand car market**
- Full-stack model: buys, inspects (200-point), refurbishes, and sells used cars
- Revenue grew from Rs. 3,262 cr (FY23) to Rs. 4,657 cr (FY25)
- Net losses cut nearly in half: Rs. 820 cr $\rightarrow$ Rs. 423 cr
- Total funding: $700 million

Key product: Assured Buyback program – guarantees a predetermined resale value for up to 18 months.

What is an Option?

A **vanilla option** gives the buyer the right, but not the obligation, to buy or sell an asset at a specific price (K) on or before a set date (T).

Put option payoff at expiration:

$$\text{Payoff} = \max(K - S_T, 0)$$

Two main types:

- **Call**: right to *buy* – exercise when $S_T > K$
- **Put**: right to *sell* – exercise when $S_T < K$

Spinny's Buyback as a Real Option

Real options apply financial option theory to business decisions.

Spinny's buyback guarantee = **Put option** for the customer:

Financial Option	**Spinny Buyback**
Underlying asset (S)	The used car
Strike price (K)	Guaranteed buyback amount
Expiration (T)	Return window (3 months)
Put payoff	$\max(K - S_T, 0)$

The customer has the right, but not the obligation, to sell the car back at K.

Option Parameters

Parameter	**Value**
Model Name	Mahindra Bolero Neo Plus P10
Model Year	2024
Selling Price (S_0)	Rs. 12,83,903
Buyback Price (K)	Rs. 11,55,641
Time to Maturity (T)	3 months
Risk-free Rate (r)	5.26% p.a.
Depreciation (δ)	10% p.a.
Volatility (σ)	15% p.a. (base case)

$K/S_0 \approx 0.90$ – the car must lose $\sim$10% for the put to be in the money.

Pricing Using Fundamentals

At 10% p.a. depreciation, the 3-month decline is:

$$1 - (0.9)^{0.25} \approx 2.60\%$$

Expected market value after 3 months:

$$S_T = 12{,}83{,}903 \times 0.974 \approx \text{Rs. } 12{,}50{,}526$$

Since $S_T \gg K$, the **intrinsic value is zero** under pure depreciation.

But if we only value the option as the discounted value of depreciation. We get, option value as $33,377 \times e^{-0.0526 \times 0.25} = 32,941$.

Binomial Model – Setup

Divide option life T into N steps of $\Delta t = T/N$. At each step:

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u}$$

Risk-neutral probability:

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

Stock price at node (i, j):

$$S_{i,j} = S_0 \cdot u^j \cdot d^{(i-j)}$$

Backward induction:

$$V_{i,j} = e^{-r\Delta t}\left[p \cdot V_{i+1,j+1} + (1-p) \cdot V_{i+1,j}\right]$$

Standard Binomial – Result

With $S_0 = \text{Rs. } 12{,}83{,}903$, $K = \text{Rs. } 11{,}55{,}641$, $\sigma = 30\%$, $N = 100$:

Up factor (u)	1.015113
Down factor (d)	0.985112
Risk-neutral p	0.502917
Put Option Value	**Rs. 20,513**
Stock + Put Value	Rs. 13,04,416

Adding Deterministic Depreciation

A used car is not a stock – it **deterministically depreciates**.

Modified stock price at each node:

$$S_{i,j} = S_0 \cdot \underbrace{(1-\delta)^{t_i}}_{\text{depreciation}} \cdot \underbrace{u^j \cdot d^{(i-j)}}_{\text{stochastic}}$$

Since depreciation is explicitly in the stock price, we use the *standard* risk-neutral probability p. (no adjustment is needed)

Over 90 days, the depreciation factor is:

$$(0.9)^{0.25} \approx 0.974 \quad \text{(base value drops } \sim 2.6\%)$$

Binomial Heatmap – Sensitivity Analysis

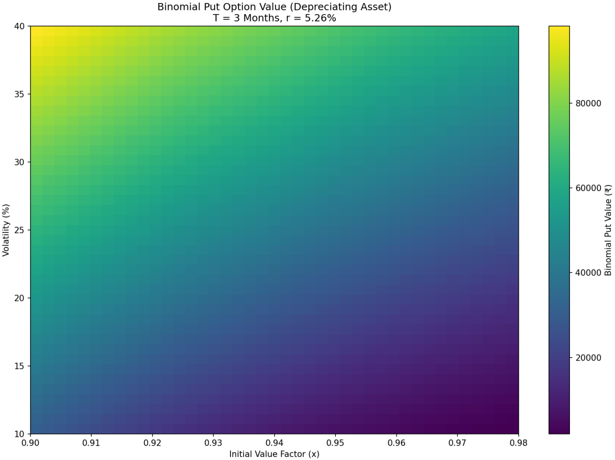

Figure: Put value across value factor (x) and volatility (σ). Range: Rs. 1,449 to Rs. 93,801.

Binomial Tree Visualisation

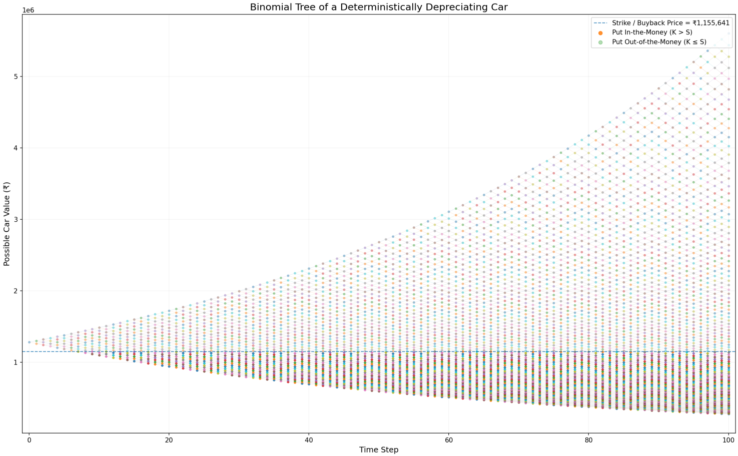

Figure: 100-period binomial tree. Brighter nodes = put in the money ($K > S$). Dashed line = buyback price.

BSM Put Formula

$$P = K\,e^{-rT}\,N(-d_2)\; - \;S_0\,N(-d_1)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)\,T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Sample values at $\sigma = 25\%$ (no depreciation):

- $S =$ Rs. 12,83,903: Put $\approx$ Rs. 12,496
- $S =$ Rs. 11,55,641 (ATM): Put $\approx$ Rs. 46,307

BSM Heatmap – No Depreciation

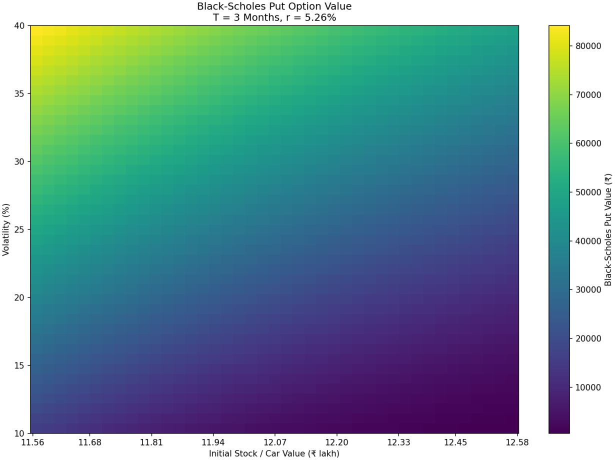

Figure: BSM put value without depreciation. Range: Rs. 385 to Rs. 80,208.

Incorporating Depreciation into BSM

Convert 10% annual depreciation to a continuous decay rate:

$$q = -\ln(1-\delta) = -\ln(0.9) \approx 10.536\% \text{ p.a.}$$

Modified BSM put formula:

$$P = K\, e^{-rT}\, N(-d_2) \, - \, S_0\, e^{-qT}\, N(-d_1)$$

$$d_1 = \frac{\ln(S_0/K) + (r - q + \frac{1}{2}\sigma^2)\, T}{\sigma\sqrt{T}}$$

Mathematically equivalent to a put on a **dividend-paying stock**, with q as the dividend yield.

BSM Heatmap – With Depreciation

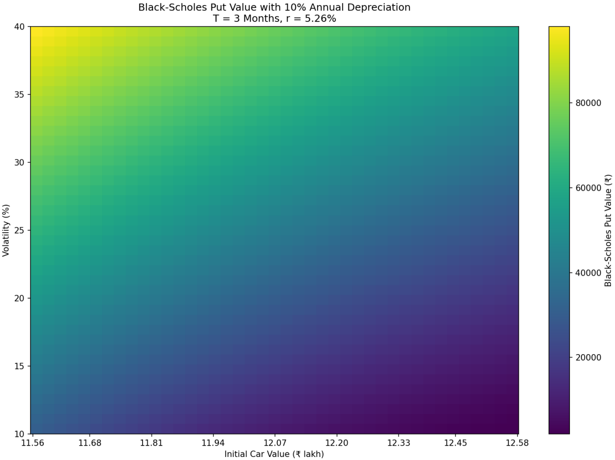

Figure: BSM put value with 10% depreciation. Range: Rs. 1,455 to Rs. 93,629. Closely matches the binomial heatmap.

BSM Values at $S_0 = \text{Rs. } 12{,}83{,}903$ (With Depreciation)

Volatility	**Put Value (Rs.)**
10%	535
20%	10,116
25%	18,139
30%	27,266
40%	47,364

At the base-case volatility of 15%, the option value is modest (∼Rs. 535). At more realistic used-car market volatilities (25–30%), it rises to Rs. 18,000–27,000.

Buyback as a Right to Abandon

In real-options theory, the right to abandon an investment = **put option** on the project value with salvage value as the strike.

For a Spinny customer:

- Buying a used car = investing in a depreciating asset
- Buyback guarantee = guaranteed exit route
- If market value drops below K, exercise the "abandonment option"

Total value of the car with the guarantee:

$$\text{Total Security Value} = S_0 + P_0$$

$$\text{Value of Embedded Put} = \text{Total Security Value} - S_0$$

Base case (standard binomial): Rs. 13,04,416 total ⇒ guarantee adds Rs. 20,513.

Consolidated Pricing

Method	Volatility	Put Value (Rs.)
Fundamental (10% dep.)	–	32,941
Standard Binomial	30%	20,513
BSM (no depreciation)	25%	12,496
BSM (with depreciation)	25%	18,139
BSM (with depreciation)	30%	27,266
BSM (with depreciation)	15%	535

Incorporating depreciation **significantly increases** the put value. At realistic volatilities (15–30%), the guarantee is worth Rs. 500 to Rs. 27,000.

How Would Spinny Price This?

Spinny's internal pricing would additionally factor in:

- Expected depreciation and buy-sell spread
- Inspection and reconditioning costs upon return
- Probability that a customer actually exercises (exercise rate)
- Cost of capital for the repurchase commitment
- Psychological value to the customer (marketing tool)

The 10% haircut ($K \approx 0.90 \times S_0$) covers expected depreciation + margin for operational costs and adverse selection.

Conclusion

- Spinny's buyback guarantee $\equiv$ **European put option** on a deterministically depreciating asset
- Three pricing methods yield consistent results: fundamental, binomial (with depreciation), and BSM (with depreciation)
- Key insight: deterministic depreciation **increases** the put value – a depreciating asset is more likely to fall below the guaranteed floor
- Option value ranges from $\sim$Rs. 500 (low σ) to $\sim$Rs. 93,000 (high σ, low x)
- Dual purpose for Spinny: marketing tool (reduces buyer anxiety) $+$ financial obligation (must be carefully priced)

Thank You

Questions?

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